

LIBREVILLE INTL, Gabon

WMO: 645000

Lat: 0.459N

Lon: 9.412E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	22.8	19.1	13.9	24.0	19.8	14.5	24.3	7.2	24.9	6.5	25.0	2.7	80	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.3	31.8	27.3	31.2	27.1	30.9	27.0	28.2	30.2	27.8	29.8	27.5	29.6	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.0	29.9	27.2	23.0	29.2	27.0	22.8	29.1	91.3	30.5	89.6	30.4	87.9	29.6	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	6.0	5.4	DB	20.7	32.9	0.9	1.4	20.1	33.9	19.6	34.7	19.1	35.5	18.4	36.5	(4)
(5)			WB	19.6	29.3	0.8	0.9	19.0	29.9	18.5	30.5	18.1	31.0	17.5	31.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	27.3	27.6	27.7	27.6	27.4	26.3	25.3	25.5	26.1	26.4	26.5	27.0	(6)	
	DBStd	1.20	0.90	1.07	0.96	0.99	0.91	0.87	0.87	0.76	0.76	0.82	0.88	0.89	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6100	537	493	548	528	539	488	473	479	484	508	496	527	(10)	
	CDD18.3	3058	278	259	290	278	281	238	214	221	234	250	246	268	(11)	
	CDH23.3	28797	2978	2922	3258	3015	2956	1989	1298	1379	1832	2171	2268	2730	(12)	
	CDH26.7	6943	856	994	1167	1010	836	248	83	95	197	326	453	679	(13)	
(14) Wind	WSAvg	3.2	2.6	2.7	2.8	2.8	3.1	3.6	3.7	3.8	3.9	3.4	3.0	2.8	(14)	
(15) Precipitation	PrecAvg	2769	262	248	357	322	246	17	2	8	116	423	496	305	(15)	
	PrecMax	3429	644	476	644	559	634	130	10	38	461	757	800	643	(16)	
	PrecMin	1977	80	78	80	10	25	0	0	0	3	0	247	59	(17)	
	PrecStd	444	121	94	138	102	154	25	3	9	114	166	155	140	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	32.1	32.2	32.1	31.4	30.0	29.1	29.1	29.4	30.1	30.5	31.0	(19)	
		MCWB	27.4	27.5	27.5	27.4	27.2	26.0	24.8	24.7	25.3	26.3	26.8	27.2	(20)	
	2%	DB	30.8	31.2	31.8	31.3	30.9	29.1	28.2	28.2	29.0	29.2	30.0	30.2	(21)	
		MCWB	27.1	27.2	27.3	27.1	27.0	25.5	24.3	24.2	25.2	25.9	26.5	26.7	(22)	
	5%	DB	30.1	30.9	31.1	31.0	30.2	28.6	27.6	27.8	28.2	29.0	29.2	29.9	(23)	
		MCWB	26.8	27.1	27.1	27.0	26.6	25.2	24.0	24.0	24.8	25.8	26.1	26.6	(24)	
	10%	DB	29.8	30.2	30.8	30.3	29.8	28.0	27.0	27.1	27.9	28.2	28.9	29.2	(25)	
		MCWB	26.6	26.7	26.9	26.7	26.5	24.8	23.6	23.6	24.7	25.4	25.9	26.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.6	28.7	28.6	28.4	26.7	25.9	25.7	26.5	27.4	27.7	28.0	(27)	
		MCDB	30.3	30.7	31.0	30.8	30.4	28.5	27.6	27.6	28.1	28.9	29.3	29.7	(28)	
	2%	WB	27.7	28.0	28.1	28.0	27.6	26.2	25.2	25.1	26.0	26.6	27.1	27.4	(29)	
		MCDB	29.5	30.1	30.3	30.0	29.6	28.0	27.0	27.0	27.7	28.2	28.7	29.1	(30)	
	5%	WB	27.2	27.6	27.6	27.5	27.2	25.7	24.7	24.7	25.6	26.2	26.7	27.0	(31)	
		MCDB	29.1	29.7	29.9	29.7	29.2	27.7	26.6	26.6	27.3	27.8	28.3	28.7	(32)	
	10%	WB	26.9	27.2	27.2	27.1	26.7	25.4	24.3	24.3	25.3	25.9	26.2	26.6	(33)	
		MCDB	28.8	29.4	29.6	29.4	28.8	27.4	26.3	26.2	27.0	27.6	27.9	28.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	6.3	6.0	5.2	4.1	4.1	4.2	4.2	4.5	5.3	5.2	(35)	
		MCDBR	5.6	6.1	6.8	6.6	5.8	4.6	4.6	4.6	4.5	5.0	5.7	5.5	(36)	
	5% WB	MCWBR	3.2	3.2	3.6	3.5	3.2	2.6	2.7	2.5	2.5	2.7	3.3	3.1	(37)	
		MCWBR	5.6	6.0	6.5	6.3	5.7	4.4	4.4	4.4	4.4	4.7	5.5	5.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.694	0.669	0.601	0.483	0.501	0.583	0.622	0.590	0.490	0.456	0.515	0.659	(40)	
		taud	1.729	1.787	1.958	2.314	2.241	1.970	1.867	1.952	2.253	2.374	2.198	1.798	(41)	
	Ebn at Noon		680	708	756	830	792	713	691	734	831	869	816	699	(42)	
		Edh at Noon	242	232	195	133	138	177	198	188	143	127	150	224	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.74	4.70	4.91	4.70	4.37	3.86	3.87	3.84	4.01	4.15	4.18	4.42	(44)	
	RadStd		0.27	0.25	0.19	0.17	0.22	0.28	0.20	0.20	0.25	0.21	0.14	0.20	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page